Ordinary Differential Equations

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September 1, 2026

Contents

1 First-Order Differential Equations **3**

1.1 Linear Differential Equations . . . 3

1.2 Separable Differential Equations . . . 4

1.3 Homogeneous Differential Equations . . . 4

2 Modeling with First-Order Equations **5**

3 Population Model **5**

3.1 Problem . . . 5

3.2 Step 1: Identify the Population . . . 6

3.3 Step 2: Model the Natural Population Growth . . . 6

3.4 Step 3: Determine the Growth Constant . . . 7

3.5 Step 4: Account for External Factors . . . 8

3.6 Step 5: Solve the Differential Equation . . . 8

3.7 Step 6: Apply the Initial Condition . . . 10

3.8 Step 7: Analyze the Population . . . 10

3.9 Equilibrium Population . . . 12

4 Mixing Problems **12**

4.1 Salt Tank Example . . . 13

4.2 Step 1: Determine the Rate of Salt Entering . . . 13

4.3 Step 2: Determine the Rate of Salt Leaving . . . 14

4.4 Step 3: Construct the Differential Equation . . . 14

4.5 Step 4: Solve the Differential Equation . . . 15

4.6 Step 5: Apply the Initial Condition 16
4.7 Step 6: Find the Limiting Amount 16

5 Key Modeling Principles **17**

5.1 General Balance Law 17
5.2 Population Models 18
5.3 Mixing Models 18
5.4 Equilibrium 18

1 First-Order Differential Equations

A first-order differential equation involves an unknown function and its first derivative. Some important types of first-order differential equations are:

- Linear equations
- Separable equations
- Homogeneous equations

1.1 Linear Differential Equations

A first-order linear differential equation has the form

$$\frac{dy}{dt} + p(t)y = g(t).$$

Equivalently,

$$\frac{dy}{dt} = -p(t)y + g(t).$$

The standard method for solving a linear equation is the **integrating factor method**. The integrating factor is

$$\mu(t) = e^{\int p(t)\,dt}.$$

Multiplying the differential equation by $\mu(t)$ gives

$$\mu(t)y' + \mu(t)p(t)y = \mu(t)g(t).$$

By construction,

$$\frac{d}{dt}\left(\mu(t)y\right) = \mu(t)y' + \mu'(t)y = \mu(t)y' + \mu(t)p(t)y.$$

Therefore,

$$\frac{d}{dt}\left(\mu(t)y\right) = \mu(t)g(t).$$

Integrating gives

$$\mu(t)y = \int \mu(t)g(t)\,dt + C.$$

Hence,

$$\boxed{y(t) = \frac{1}{\mu(t)} \left(\int \mu(t) g(t)\, dt + C \right).}$$

1.2 Separable Differential Equations

A differential equation is **separable** if it can be written in the form

$$\frac{dy}{dx} = \frac{M(x)}{N(y)}.$$

Rearranging gives

$$N(y)\, dy = M(x)\, dx.$$

Integrating both sides gives

$$\int N(y)\, dy = \int M(x)\, dx.$$

This generally produces an **implicit solution**.

Thus, the basic procedure is

$$\boxed{\frac{dy}{dx} = \frac{M(x)}{N(y)} \quad \Longrightarrow \quad N(y)\, dy = M(x)\, dx \quad \Longrightarrow \quad \int N(y)\, dy = \int M(x)\, dx.}$$

1.3 Homogeneous Differential Equations

A first-order differential equation is homogeneous if it can be written in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

The standard substitution is

$$v = \frac{y}{x}.$$

Since

$$y = xv,$$

differentiating with respect to x gives

$$\frac{dy}{dx} = v + x\frac{dv}{dx}.$$

Substituting this into the original differential equation gives an equation involving v and x. This equation is often separable.

Thus,

$$\boxed{v = \frac{y}{x}}$$

transforms a first-order homogeneous equation into a separable equation.

2 Modeling with First-Order Equations

Differential equations are frequently used to construct mathematical models of real-world systems.

A general modeling process is:

1. **Construct the model.**
2. **Analyze the model.**
3. **Compare the model with data and/or make predictions.**

The first step is to identify the quantity being modeled.

For example, if $A(t)$ represents the amount of some quantity at time t, then its rate of change can often be expressed as

$$\boxed{\frac{dA}{dt} = \text{sum of gain rates} - \text{sum of loss rates.}}$$

This principle is especially useful for population models and mixing problems.

3 Population Model

3.1 Problem

An insect population in a region will increase at a rate proportional to its current size. In the absence of any external factors, the population triples in two weeks.

On any given day, there is a net migration of 15 insects into the region, 16 insects are eaten by birds, and 7 insects die of natural causes.

If initially there are 100 insects in the area, set up a differential equation model to study whether this population will survive. If not, estimate when the insects die out.

3.2 Step 1: Identify the Population

Let

$$p(t)$$

denote the insect population at time t, measured in days.

The initial condition is

$$\boxed{p(0) = 100.}$$

3.3 Step 2: Model the Natural Population Growth

The statement

"population increases at a rate proportional to its current size"

means

$$\frac{dp}{dt} = rp,$$

where

$$r > 0$$

is a constant of proportionality.

This is the standard exponential growth model.

Separating variables gives

$$\frac{1}{p}\,dp = r\,dt.$$

Integrating,

$$\int \frac{1}{p}\,dp = \int r\,dt,$$

so

$$\ln|p| = rt + C.$$

Exponentiating,

$$p = Ce^{rt}.$$

Using $p(0) = p_0$,

$$p_0 = Ce^0 = C.$$

Therefore,

$$\boxed{p(t) = p_0 e^{rt}.}$$

3.4 Step 3: Determine the Growth Constant

In the absence of external factors, the population triples in two weeks.

Since t is measured in days,

$$p(14) = 3p(0).$$

Using the exponential model,

$$p(14) = p_0 e^{14r}.$$

Therefore,

$$p_0 e^{14r} = 3p_0.$$

Canceling p_0,

$$e^{14r} = 3.$$

Taking the natural logarithm,

$$14r = \ln(3).$$

Thus,

$$\boxed{r = \frac{\ln(3)}{14}.}$$

3.5 Step 4: Account for External Factors

There are three external effects:

- 15 insects enter the region per day.
- 16 insects are eaten by birds per day.
- 7 insects die naturally per day.

The total gain rate is

$$15.$$

The total loss rate is

$$16 + 7 = 23.$$

Therefore, the net external rate is

$$15 - 23 = -8.$$

Thus,

$$\boxed{\frac{dp}{dt} = rp - 8.}$$

Equivalently,

$$\boxed{\frac{dp}{dt} - rp = -8.}$$

This is a first-order linear differential equation.

3.6 Step 5: Solve the Differential Equation

The equation is

$$p' - rp = -8.$$

Comparing this with

$$p' + a(t)p = g(t),$$

we have

$$a(t) = -r.$$

Therefore, the integrating factor is

$$\mu(t) = e^{\int -r\,dt} = e^{-rt}.$$

Multiplying the differential equation by e^{-rt},

$$e^{-rt}p' - re^{-rt}p = -8e^{-rt}.$$

The left-hand side is

$$\frac{d}{dt}\left(e^{-rt}p\right).$$

Therefore,

$$\frac{d}{dt}\left(e^{-rt}p\right) = -8e^{-rt}.$$

Integrating,

$$e^{-rt}p = \int -8e^{-rt}\,dt + C.$$

Since

$$\int -8e^{-rt}\,dt = \frac{8}{r}e^{-rt},$$

we obtain

$$e^{-rt}p = \frac{8}{r}e^{-rt} + C.$$

Multiplying by e^{rt},

$$p(t) = \frac{8}{r} + Ce^{rt}.$$

Thus, the general solution is

$$\boxed{p(t) = \frac{8}{r} + Ce^{rt}.}$$

3.7 Step 6: Apply the Initial Condition

We know that

$$p(0) = 100.$$

Therefore,

$$100 = \frac{8}{r} + C.$$

Hence,

$$C = 100 - \frac{8}{r}.$$

Since

$$r = \frac{\ln(3)}{14},$$

we have

$$\frac{8}{r} = \frac{8(14)}{\ln(3)} = \frac{112}{\ln(3)} \approx 101.95.$$

Therefore,

$$C \approx 100 - 101.95 = -1.95.$$

The population function is therefore

$$\boxed{p(t) = 101.95 - 1.95e^{(\ln(3)/14)t}.}$$

3.8 Step 7: Analyze the Population

Because

$$r = \frac{\ln(3)}{14} > 0,$$

we have

$$e^{rt} \to \infty \qquad \text{as } t \to \infty.$$

Since the coefficient of the exponential term is negative,

$$-1.95e^{rt} \to -\infty.$$

Therefore,

$$\boxed{p(t) \to -\infty \qquad \text{as } t \to \infty.}$$

Of course, a negative population is not physically meaningful. Instead, the model tells us that the population reaches zero at some finite time.

We therefore solve

$$p(t^*) = 0.$$

Thus,

$$101.95 - 1.95e^{(\ln(3)/14)t^*} = 0.$$

Rearranging,

$$1.95e^{(\ln(3)/14)t^*} = 101.95,$$

so

$$e^{(\ln(3)/14)t^*} = \frac{101.95}{1.95}.$$

Taking the natural logarithm,

$$\frac{\ln(3)}{14}t^* = \ln\left(\frac{101.95}{1.95}\right).$$

Therefore,

$$\boxed{t^* = \frac{14}{\ln(3)}\ln\left(\frac{101.95}{1.95}\right).}$$

Numerically,

$$\boxed{t^* \approx 40.2 \text{ days.}}$$

Thus, according to this model, the population dies out after approximately 40 days.

3.9 Equilibrium Population

An equilibrium population occurs when

$$\frac{dp}{dt} = 0.$$

For

$$\frac{dp}{dt} = rp - 8,$$

we set

$$rp - 8 = 0.$$

Therefore,

$$p^* = \frac{8}{r}.$$

Using

$$r = \frac{\ln(3)}{14},$$

we obtain

$$\boxed{p^* = \frac{112}{\ln(3)} \approx 101.95.}$$

This equilibrium is important because it determines the long-term behavior of the model.

Since

$$p(0) = 100 < 101.95 = p^*,$$

the population begins below equilibrium.

Because $r > 0$, populations below the equilibrium decrease toward zero and eventually reach zero in the physical interpretation of the model.

4 Mixing Problems

Mixing problems model the amount of a substance in a container when a solution enters and leaves the container.

The central idea is again

$$\boxed{\frac{dQ}{dt} = \text{rate in} - \text{rate out},}$$

where $Q(t)$ is the amount of substance in the tank.

For a salt solution,

$$\boxed{\frac{dQ}{dt} = (\text{salt concentration in})(\text{flow rate in}) - (\text{salt concentration out})(\text{flow rate out}).}$$

4.1 Salt Tank Example

At $t = 0$, a tank contains 100 gallons of water.

A solution containing

$$\frac{1}{4} \text{ lb of salt per gallon}$$

enters the tank at a rate of r gallons per minute.

The tank is well stirred, and the solution leaves the tank at the same rate r gallons per minute.

Let

$$Q(t)$$

denote the amount of salt in the tank at time t, measured in pounds.

Suppose

$$Q(0) = Q_0.$$

We want to:

1. Find $Q(t)$, the amount of salt in the tank at any time.
2. Find the limiting amount of salt in the tank.

4.2 Step 1: Determine the Rate of Salt Entering

The incoming solution has concentration

$$\frac{1}{4}\frac{\text{lb}}{\text{gal}}.$$

The flow rate is

$$r\frac{\text{gal}}{\text{min}}.$$

Therefore,

$$\text{rate in} = \left(\frac{1}{4}\frac{\text{lb}}{\text{gal}}\right)\left(r\frac{\text{gal}}{\text{min}}\right).$$

The gallons cancel, leaving

$$\boxed{\text{rate in} = \frac{r}{4}\frac{\text{lb}}{\text{min}}.}$$

4.3 Step 2: Determine the Rate of Salt Leaving

Because the tank is well stirred, the concentration of salt leaving the tank is the same as the concentration inside the tank.

At time t, the tank contains $Q(t)$ pounds of salt and 100 gallons of solution.

Therefore, the concentration inside the tank is

$$\boxed{\frac{Q(t)}{100}\frac{\text{lb}}{\text{gal}}.}$$

The solution leaves at a rate of

$$r\frac{\text{gal}}{\text{min}}.$$

Thus,

$$\text{rate out} = \left(\frac{Q(t)}{100}\frac{\text{lb}}{\text{gal}}\right)\left(r\frac{\text{gal}}{\text{min}}\right).$$

Therefore,

$$\boxed{\text{rate out} = \frac{rQ}{100}\frac{\text{lb}}{\text{min}}.}$$

4.4 Step 3: Construct the Differential Equation

Using

$$\frac{dQ}{dt} = \text{rate in} - \text{rate out},$$

we obtain

$$\frac{dQ}{dt} = \frac{r}{4} - \frac{rQ}{100}.$$

Therefore,

$$\boxed{\frac{dQ}{dt} = \frac{r}{4} - \frac{r}{100}Q.}$$

Equivalently,

$$\boxed{\frac{dQ}{dt} + \frac{r}{100}Q = \frac{r}{4}.}$$

This is a first-order linear differential equation.

4.5 Step 4: Solve the Differential Equation

The equation has the form

$$Q' + p(t)Q = g(t),$$

where

$$p(t) = \frac{r}{100}$$

and

$$g(t) = \frac{r}{4}.$$

The integrating factor is

$$\mu(t) = e^{\int (r/100)\,dt} = e^{rt/100}.$$

Multiplying through by the integrating factor,

$$e^{rt/100}Q' + \frac{r}{100}e^{rt/100}Q = \frac{r}{4}e^{rt/100}.$$

The left-hand side becomes

$$\frac{d}{dt}\left(e^{rt/100}Q\right).$$

Therefore,

$$\frac{d}{dt}\left(e^{rt/100}Q\right) = \frac{r}{4}e^{rt/100}.$$

Integrating,

$$e^{rt/100}Q = \int \frac{r}{4}e^{rt/100}\,dt + C.$$

Since

$$\int \frac{r}{4}e^{rt/100}\,dt = 25e^{rt/100},$$

we obtain

$$e^{rt/100}Q = 25e^{rt/100} + C.$$

Thus,

$$\boxed{Q(t) = 25 + Ce^{-rt/100}.}$$

4.6 Step 5: Apply the Initial Condition

Since

$$Q(0) = Q_0,$$

we have

$$Q_0 = 25 + C.$$

Therefore,

$$C = Q_0 - 25.$$

The solution is

$$\boxed{Q(t) = 25 + (Q_0 - 25)e^{-rt/100}.}$$

4.7 Step 6: Find the Limiting Amount

To find the limiting amount, consider

$$\lim_{t\to\infty} Q(t).$$

Since $r > 0$,

$$e^{-rt/100} \to 0 \qquad \text{as } t \to \infty.$$

Therefore,

$$\lim_{t\to\infty} Q(t) = 25.$$

Hence,

$$\boxed{\lim_{t\to\infty} Q(t) = 25 \text{ lb}.}$$

This makes sense physically because the incoming solution has concentration

$$\frac{1}{4} \text{ lb/gal},$$

and the tank contains 100 gallons. Therefore, at equilibrium,

$$Q^* = \left(\frac{1}{4}\frac{\text{lb}}{\text{gal}}\right)(100 \text{ gal}) = 25 \text{ lb}.$$

Thus, the equilibrium amount is exactly the amount corresponding to the concentration of the incoming solution.

5 Key Modeling Principles

The examples above illustrate several important principles.

5.1 General Balance Law

For many models,

$$\boxed{\text{Rate of change} = \text{Rate in} - \text{Rate out}.}$$

More generally,

$$\boxed{\frac{dA}{dt} = \text{sum of gain rates} - \text{sum of loss rates}.}$$

5.2 Population Models

A population with growth proportional to its current size satisfies

$$\boxed{p' = rp.}$$

If there is also a constant net external contribution k, then

$$\boxed{p' = rp + k.}$$

5.3 Mixing Models

For a well-stirred tank,

$$\boxed{Q' = (\text{incoming concentration})(\text{incoming flow rate}) - (\text{tank concentration})(\text{outgoing flow rate}).}$$

The key assumption is that the tank is **well stirred**, so the concentration of the material leaving the tank is equal to the concentration throughout the tank.

5.4 Equilibrium

An equilibrium occurs when the quantity is not changing:

$$\boxed{\frac{dA}{dt} = 0.}$$

For an equation

$$A' = f(A),$$

an equilibrium A^* satisfies

$$\boxed{f(A^*) = 0.}$$

Equilibria are useful for understanding the long-term behavior of solutions without necessarily solving the differential equation.